

AUTOSAR HLD Intelligence & Traceability Platform

Technical Report

Case Study CS1: AUTOSAR HLD Document Analysis Assistant
TechPulse FY-26 | Applied AI/ML Capstone

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Case study	CS1: AUTOSAR HLD Document Analysis Assistant
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All input data is synthetic. AI-generated outputs are advisory and require review by a qualified engineer.

Summary and map to the evaluation rubric

This report describes a retrieval-augmented assistant for AUTOSAR High-Level Design (HLD) documents. It extracts the architecture (components, interfaces, ports, signals, dependencies, flows), answers questions with page citations, refuses questions the document cannot answer, validates cross-table consistency, compares revisions and exports structured findings. Its learned component is an **answerability classifier** selected from six model families by nested leave-one-document-out cross-validation (random forest selected; held-out test F1 0.96 vs 0.73 for the earlier hand-set threshold).

Rubric criterion	Where addressed
1 Problem understanding and context	Section 1 (problem, users, measurable success criteria)
2 AI/ML fundamentals and approach selection	Section 4 (alternatives compared with evidence)
3 Dataset understanding, preprocessing, EDA	Sections 2 and 3 (corpus, splits, EDA, leakage checks)
4 Feature engineering, selection, dimensionality reduction	Section 6 (tokenisation ablation, 12 features, importance, PCA)
5 Model development with taught techniques	Section 7 (retrieval models, six classifier families, tuning)
6 Evaluation, validation, optimisation	Section 8 (nested CV, overfitting, errors, held-out and independent tests)
7 Completeness and working demonstration	Sections 5, 9, 13 (end-to-end system, screenshots)
8 Python implementation and code quality	Section 9 (modules, tests, linting)
9 Deployment and MLOps basics	Section 10 (versioning, tracking, Docker, CI)
10 Responsible AI and ethics	Section 11
11-12 Technical Q&A, communication	Section 14 and the Technical Q&A preparation notes

1. Problem statement, users and success criteria

AUTOSAR HLD documents describe software components, interfaces, ports, signals, dependencies and functional flows. They are long unstructured PDFs, so architects and integrators read them manually to find mismatches and missing dependencies and to judge what a change affects. Manual review is slow, inconsistent and hard to audit.

Intended users: system and AUTOSAR architects, software developers, integration engineers, test engineers and engineering managers. **Out of scope:** automatic approval of architecture decisions and modification of source HLDs; every output is advisory and passes a human review step.

Measurable success criteria (defined before the experiments, results measured on held-out data):

Criterion	Target	Measured	Where
Entity extraction F1 on unseen documents	≥ 0.95	1.00	7.3
Refusal accuracy, independent hand-written set	≥ 0.90	0.92	8.4
Answerable-question accuracy, independent set	≥ 0.85	0.88	8.4
Answerability F1, unseen documents	≥ 0.90	0.96	8.3
Answers carrying valid citations	100%	100.0%	8.4
Seeded-defect recall / false positives on clean documents	$\geq 0.90 / 0$	1.00 / 0	8.6

Business value: less manual review effort, a searchable architecture repository, traceability and impact analysis, and an auditable record of which findings humans accepted or rejected.

2. Data and knowledge base

No real HLD was available, and real ones are confidential, so the project uses **synthetic HLDs** generated with ReportLab in the layout of the sample document (system overview, components, interfaces, ports, signals, dependency table, functional flows, constraints). Eleven documents are used:

Group	Documents	Use
Development (6)	powertrain, adas, infotainment, battery, lighting, thermal (4-6 components each)	Developing extractors, training and cross-validating the classifier, tuning the retrieval weight
Held-out test (3)	steering, keyless, wiper (generated after the development set was frozen)	Final evaluation only
Sample + revision (2)	sample_hld.pdf (Body Control System) and sample_hld_v2.pdf	Demo document, revision comparison, hand-written question set, final test

Each synthetic document has a **ground-truth file** with its entities, and a **defective twin** with four seeded defects (wrong port direction, dropped dependency, undefined interface on a port, signal endpoint mismatch). **Question data:** for every document, 24 answerable questions are generated from its own tables (so gold answers and gold sections are known) and 24 unanswerable ones of three kinds: off-topic, plausible-but-absent attributes of real entities (for example the CAN baud rate of a component), and invented entities. This gives 528 labelled questions (288 development, 240 test), balanced 50/50. Two **hand-written** sets provide independent checks: 32 questions on the sample HLD and 36 questions on the three held-out domains, written before the final model was evaluated.

Limitation: the data is synthetic and uniform in layout. Results show that the method works and generalises across domains and naming styles; they do not establish accuracy on real-world HLDs.

3. Exploratory data analysis and preprocessing

The corpus has 11 documents, 35 pages, 4,614 words, 55 tables and 121 section-aware chunks (mean 38.1 words, maximum 136). The most relevant observations:

- **Table text artefacts:** 203 table cells contain a PDF line-wrap artefact in which a long identifier such as DoorStatus_Out is extracted as 'DoorStatus Out' followed by a stray underscore line. Left untreated, port names would not match their definitions. A cleaning step rejoins them (preprocessing).
- **OCR trigger:** the first version sent any page with fewer than 20 words to OCR, including nearly empty text pages at the end of a section. OCR is now used only for pages with images or no text layer; no page in the corpus needs it.
- **Chunk size:** chunks are short (median 30 words) because HLD sections are short tables, so the 250-word chunk limit rarely applies here. It is set to stay within the 512-token window of the embedding model; with 1200 words long sections would be silently truncated.
- **Missing values:** 0 in the feature matrix; 1 duplicate question row (negligible).
- **Class balance:** 144 / 144 (development) and 120 / 120 (test) answerable vs unanswerable, so accuracy is meaningful; F1 and ROC-AUC are reported as well.
- **Leakage control:** no document appears in both development and test sets. 30 identical questions occur in both splits (15.0% of the test set); these are generic off-topic sentences that are unanswerable in every document, so they cannot leak document-specific information. Templates are shared by design.
- **Feature relationships:** several features are strongly correlated (top-1/top-2/mean similarity, 0.97-0.99; BM25 score vs score per token, 0.94; coverage vs uncovered tokens, -0.85). This motivates regularised and tree-based models and the PCA analysis in Section 6.

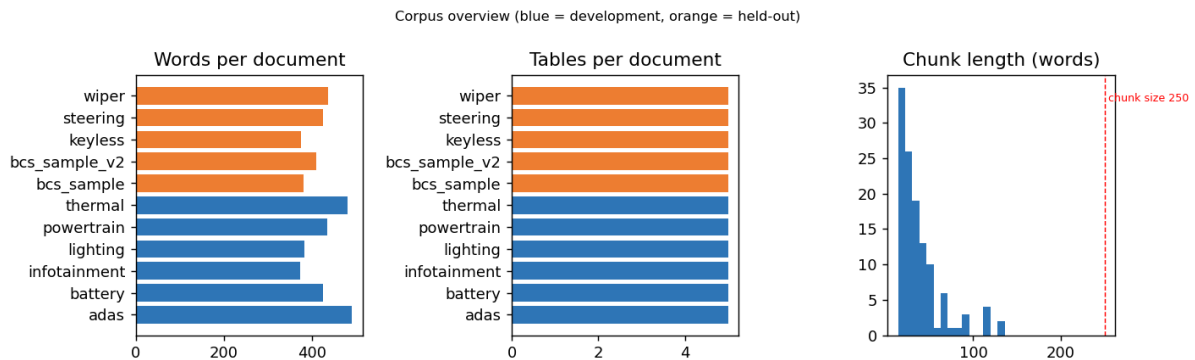


Figure 1: Corpus overview and chunk-length distribution.

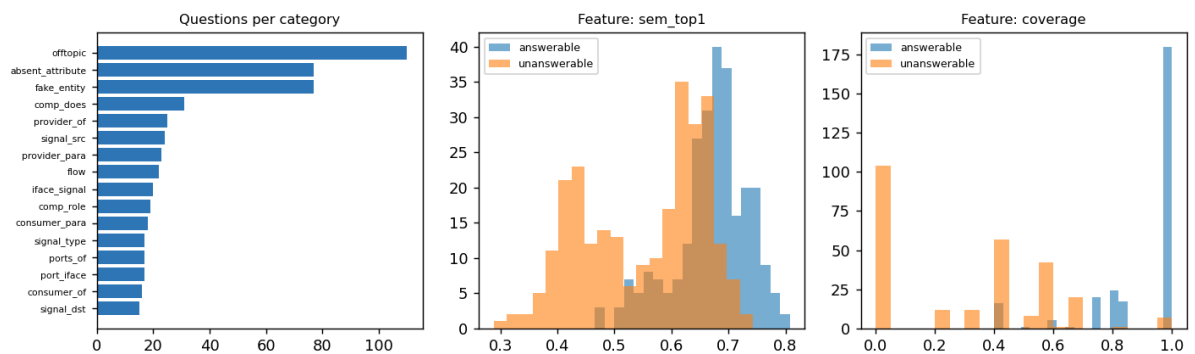


Figure 2: Question categories and two discriminative features by class.

4. Approach selection and justification

Three design questions were answered by comparing alternatives rather than by assumption.

Decision	Alternatives	Choice and evidence
How to extract architecture entities	(a) generative LLM; (b) naming-pattern regular expressions only; (c) table parsing plus patterns	(c). Patterns alone reached F1 0.83 on development domains (signal recall 0.47, flow recall 0): names that do not end in Status/Speed/Position/Command were missed. Adding table-driven extraction and parsing titles from structure gave F1 1.00 on three further unseen domains. Deterministic extraction keeps every entity traceable to a page and is reproducible; an LLM would add cost and non-determinism.
How to retrieve evidence	BM25, TF-IDF, MiniLM embeddings, BGE embeddings, hybrid	Weighted hybrid of BGE dense retrieval and BM25 (held-out MRR 0.869). BGE alone scored 0.865; BM25 0.828; TF-IDF 0.799; MiniLM 0.787. See Section 7.
When to refuse a question	(a) hand-set similarity threshold; (b) trained classifier	(b). The 0.60 threshold gave held-out F1 0.73; it answered 83.3% of the unanswerable questions in the independent set. The trained classifier reached F1 0.96 (Section 8).
How to phrase answers	LLM generation vs extractive composition	Extractive composition from retrieved passages and graph facts by default: no hallucination risk, offline, auditable. An optional local LLM (Ollama) can rewrite the evidence; it was not part of the evaluation.

Learning types. Embedding models are pre-trained (self-supervised); the answerability classifier is **supervised binary classification** trained on labelled questions; retrieval ranking and extraction are unsupervised/rule-based. The task has a small labelled corpus, so low-variance models and grouped cross-validation are appropriate.

5. Solution architecture

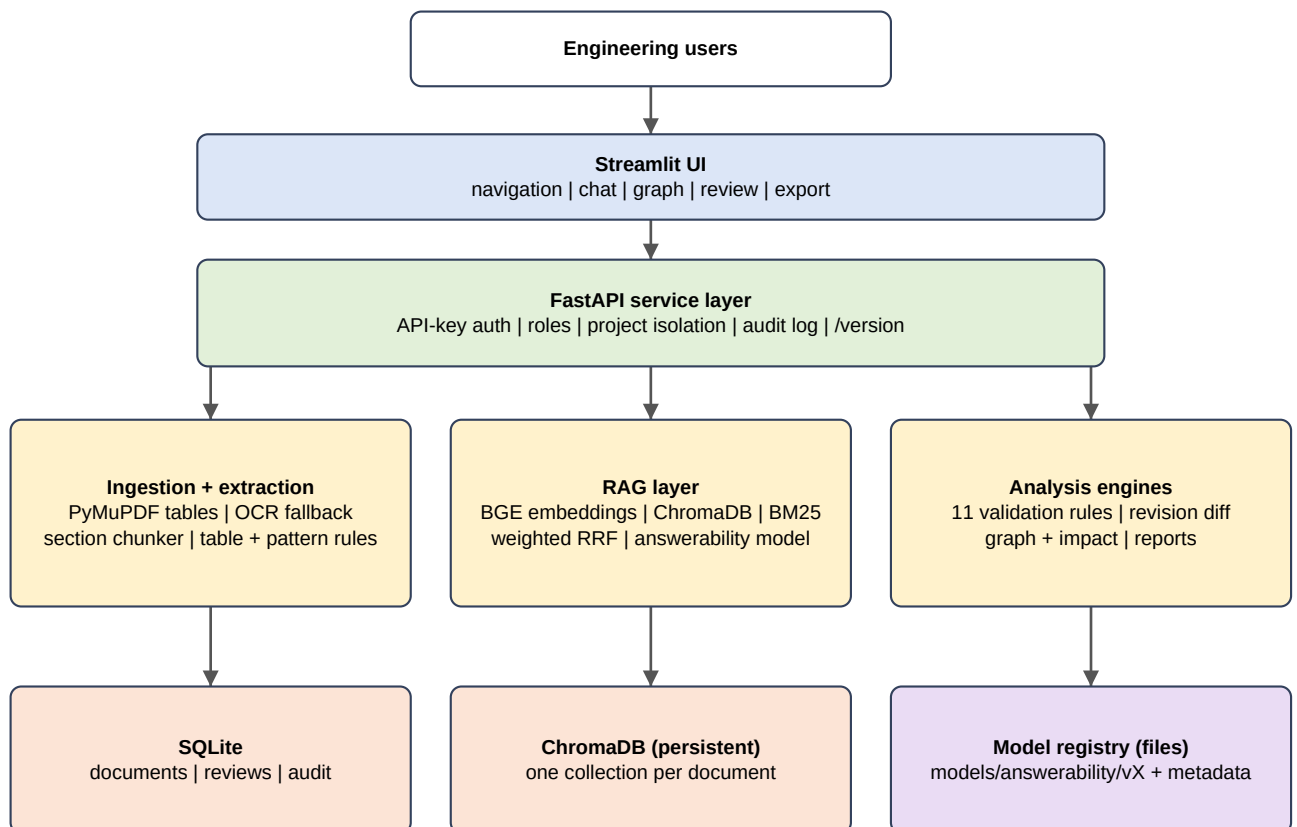


Figure 3: System architecture.

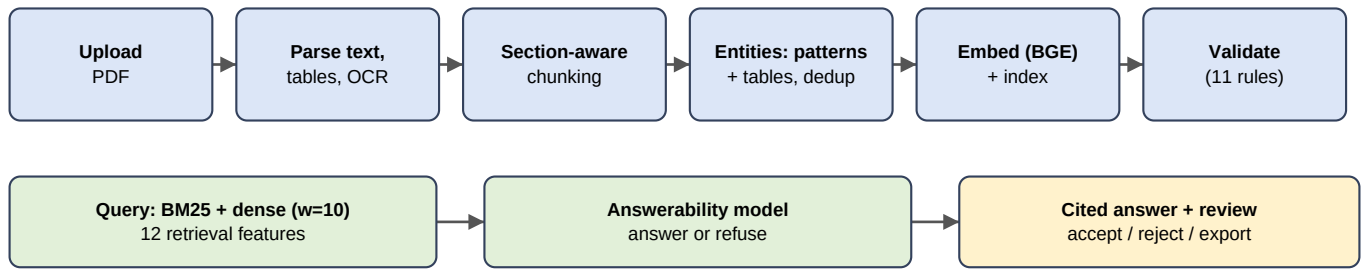


Figure 4: Processing and query workflow.

Layer	Choice	Note
UI / API	Streamlit / FastAPI	As recommended in the case-study document
Ingestion	PyMuPDF (text, tables), Tesseract OCR for scanned pages	Table cleaning for wrapped identifiers
Embeddings / vectors	BAAI/bge-small-en-v1.5 (ONNX, local), ChromaDB persistent, one collection per document	Runs offline from the Docker image
Structured store	SQLite: documents, reviews, audit log	PostgreSQL for production
Graph	In-memory property graph built from the HLD tables	Neo4j not used at this document size
LLM	Off by default; optional Ollama (local) or Anthropic (external, must be declared)	Not part of reported results

6. Feature engineering, selection and dimensionality reduction

6.1 Text representation

HLD identifiers are CamelCase (DoorControl) and snake_case (DoorStatus_In) so a plain tokeniser cannot match the natural-language question 'door control' to the identifier. The tokeniser indexes each identifier whole and split into words, and drops function words (a stop-word list extended after error analysis showed that words such as 'have' counted as unseen vocabulary). Ablation: BM25 with plain tokens reached held-out MRR 0.817; the CamelCase-aware tokeniser reached 0.828 (recall@1 0.717 to 0.733). The gain is modest because the dense model also handles part of the matching.

6.2 Answerability features

Feature	Meaning
sem_top1, sem_top2, sem_margin, sem_mean5	Cosine similarity of the best, second-best chunk, their gap, mean of top five
bm25_top1, bm25_per_token	Best BM25 score, and per query token
coverage, uncovered, n_tokens	Share and count of query words that exist anywhere in the document vocabulary; query length
agree	1 if BM25 and dense retrieval choose the same top chunk
top_chunk_cov	Share of query words found in the best chunk
entity_match	Number of known architecture entities named in the question

6.3 Feature selection and importance

Permutation importance on the held-out documents and leave-one-feature-out ablation (logistic regression) were used. Using all features gives test AUC 0.9876; removing any single feature changes it by at most 0.013, which confirms the redundancy seen in the correlation matrix. The most informative features are those that measure how much of the question the document actually contains (coverage, entity match, similarity) rather than the raw BM25 magnitude.

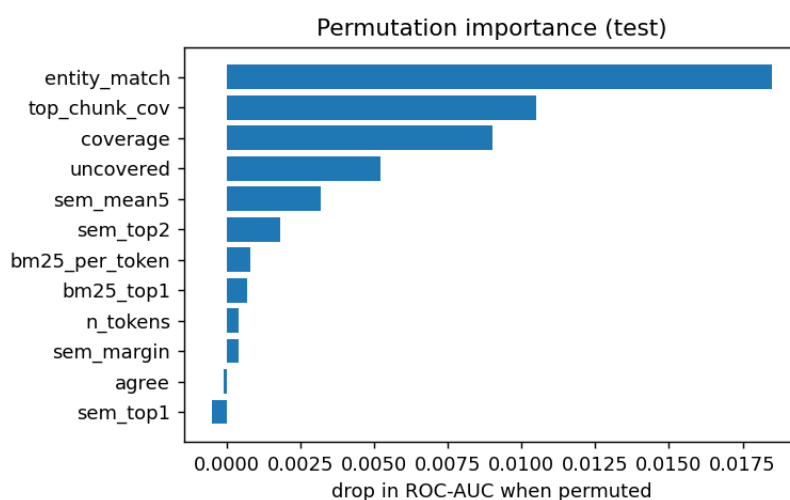


Figure 5: Permutation importance on held-out documents.

6.4 Dimensionality reduction (PCA)

PCA on the 12 standardised features: the first 4 components explain at least 90% of the variance. A logistic regression trained on PCA scores recovers most of the full-feature AUC with few components (figure), so PCA is a valid compression but gives no accuracy gain; the classifier uses the original, interpretable features. PCA was also applied to the 384-dimensional chunk embeddings: reducing to 64 dimensions lowered held-out retrieval MRR from 0.865 to 0.771 and to 0.741 at 16 dimensions, so the full embedding is kept. (With only 66 development chunks, components beyond about 30 are noise; this limits the experiment.)

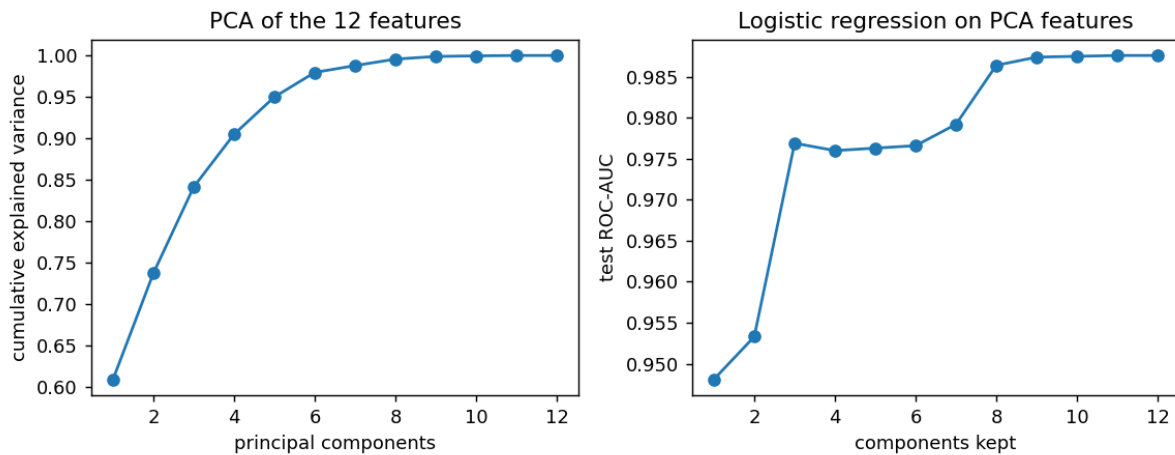


Figure 6: PCA of the answerability features and downstream AUC.

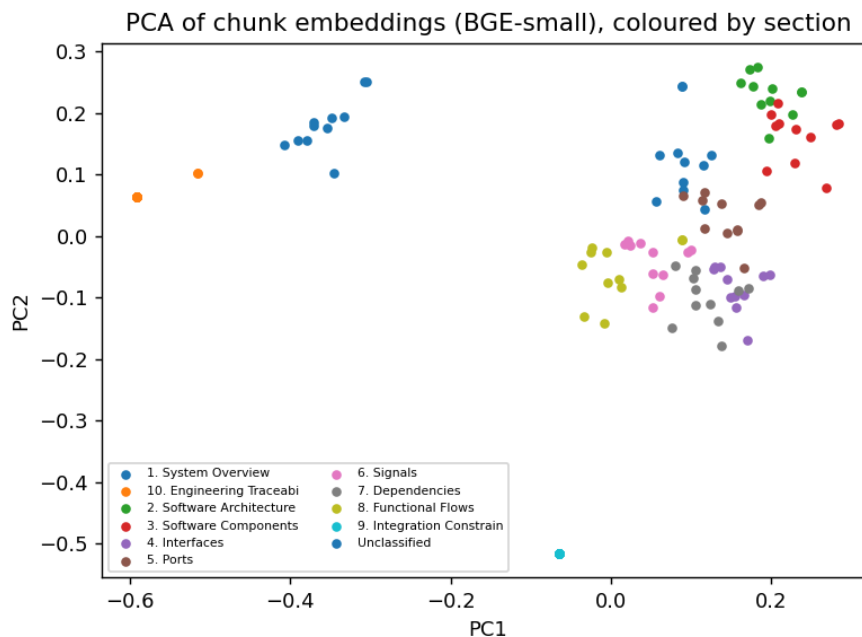


Figure 7: PCA of BGE chunk embeddings, coloured by HLD section.

7. Model development

7.1 Retrieval models

Retrieval ranks the chunks of a document for each answerable question; a result is correct if a chunk from a gold section is returned. Taught techniques compared: lexical (BM25, TF-IDF), two pre-trained sentence-embedding models and hybrids.

Retrieval method	Dev MRR	Test R@1	Test R@3	Test MRR
BM25 (plain tokens)	0.805	0.717	0.892	0.817
BM25 (CamelCase-aware, used)	0.8	0.733	0.892	0.828
TF-IDF cosine	0.809	0.692	0.875	0.799
Dense: MiniLM-L6	0.822	0.642	0.942	0.787
Dense: BGE-small (used)	0.858	0.792	0.925	0.865
Hybrid BM25 + BGE, equal weights	0.822	0.717	0.95	0.837
Hybrid BM25 + BGE, RRF dense weight 10.0 (tuned on dev)	0.863	0.792	0.942	0.869

The equal-weight hybrid was worse than BGE alone, which contradicted the initial design. The dense weight in the reciprocal-rank fusion was therefore treated as a hyperparameter and tuned on the development documents only (grid 1.0, 1.5, 2.0, 3.0, 5.0, 10.0); weight 10.0 won and was confirmed on held-out documents (MRR 0.869). BM25 now mostly acts as a tie-breaker and keeps exact identifiers findable.

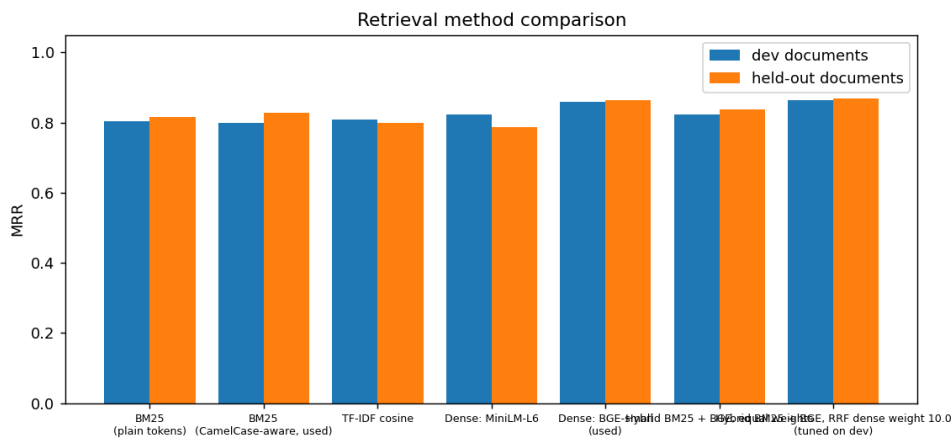


Figure 8: Retrieval comparison on development and held-out documents.

7.2 Answerability classifier (supervised learning)

Task: given a question and the retrieval evidence from one document, predict whether the document can answer it. Six taught model families were compared with hyperparameter grids: logistic regression (C), RBF-SVM (C, gamma), k-nearest neighbours (k), decision tree (depth), random forest (depth, leaf size) and gradient boosting (trees, depth, learning rate), plus two baselines: the previous hand-set threshold and the majority class. Features are standardised for the distance- and margin-based models. **Protocol:** nested cross-validation grouped by *document* (outer leave-one-document-out, inner 5-fold grouped grid search), so every score is for a document the model has never seen and hyperparameters are not tuned on the evaluation fold. The decision threshold was chosen on out-of-fold predictions.

Model	CV AUC (mean $\pm$ std)	Train AUC	Overfit gap	F1 (tuned thr.)
rule: sem_top1 $\geq$ 0.60 (previous)	0.834 $\pm$ 0.000	-	-	0.7477

Model	CV AUC (mean $\pm$ std)	Train AUC	Overfit gap	F1 (tuned thr.)
majority class	0.500 $\pm$ 0.000	-	-	0.667
logistic_regression	0.988 $\pm$ 0.014	0.9942	0.0061	0.9619
svm_rbf	0.991 $\pm$ 0.009	0.9976	0.0066	0.9863
knn	0.992 $\pm$ 0.010	0.9979	0.006	0.9795
decision_tree	0.964 $\pm$ 0.020	0.9861	0.0224	0.9408
random_forest	0.994 $\pm$ 0.007	0.9999	0.0063	0.9795
gradient_boosting	0.988 $\pm$ 0.016	0.9998	0.0123	0.9793

All learned models clearly beat the hand-set rule (AUC 0.83) and the majority baseline. Differences among the top models (SVM, k-NN, random forest, boosting) are within one standard deviation, so the choice is not critical; **random forest** was selected for the highest mean AUC and wrapped in sigmoid calibration so its probability can serve as a confidence value. Chosen hyperparameters: {'max_depth': 'None', 'min_samples_leaf': 1}; decision threshold 0.35.

7.3 Extraction improvements found by error analysis

Running the first extractors on the six development domains exposed generalisation failures. They were fixed one cause at a time, then confirmed on three further domains that had not been used for debugging.

Failure (before)	Cause	Fix
Signal recall 0.47	Signals recognised only if named *Status/Speed/Position/Command	Read signals, components, interfaces and ports directly from the tables
Flow precision/recall 0/0	Flow titles taken from a hard-coded list of the sample document	Title = leading plain words before the first identifier-like token
Interface precision 0.94	Pattern matched all-caps words such as IVI	Require a lowercase letter after the leading I
OCR attempted on blank text pages	Low word count treated as scanned	OCR only for pages with images or no text layer

Split	Overall precision	Overall recall	F1
Development, before fixes	0.86	0.81	0.834
Development, after fixes	1.0	1.0	1.0
Held-out (3 unseen domains), after fixes	1.0	1.0	1.0

Honest note: the development numbers after the fixes are optimistic because the fixes were derived from those documents; the held-out row is the fair estimate, and it is perfect partly because the synthetic documents share one table layout.

8. Evaluation, validation and optimisation

8.1 Protocol

Development documents were used for model development, cross-validation and tuning. The held-out documents (three unseen domains plus the sample HLD and its revision) were evaluated once, after the model, threshold and retrieval weight were fixed. Metrics: ROC-AUC (threshold-free), precision, recall, F1 for the answerable class, accuracy, Brier score (calibration), and for retrieval recall@k and MRR.

8.2 Validation and overfitting analysis

Nested cross-validation AUC for random forest is 0.994 ± 0.007 across six unseen documents, against a training AUC of 0.9999; the gap of 0.0063 is small, and the held-out test AUC (0.994) is in line with the cross-validation estimate, which indicates no material overfitting. The decision tree, the least regularised model, has the largest gap-to-score ratio and the lowest AUC. The learning curve shows validation AUC rising and flattening with more training examples, so more documents would help only marginally.

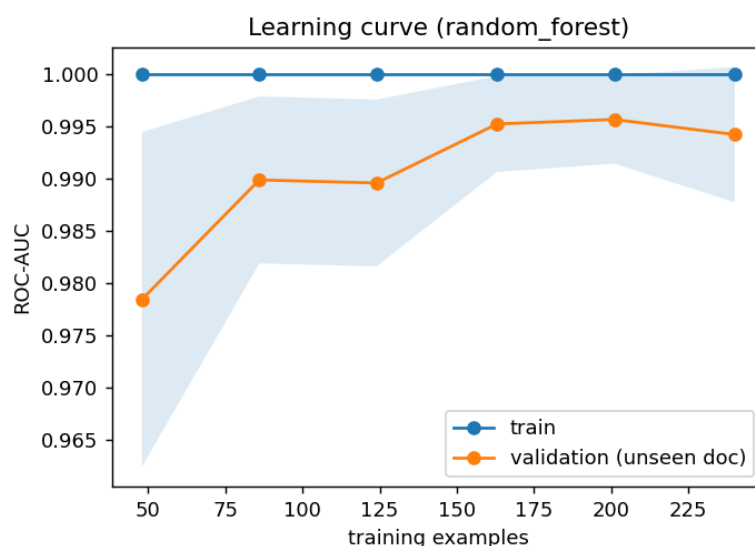


Figure 9: Learning curve (grouped by document).

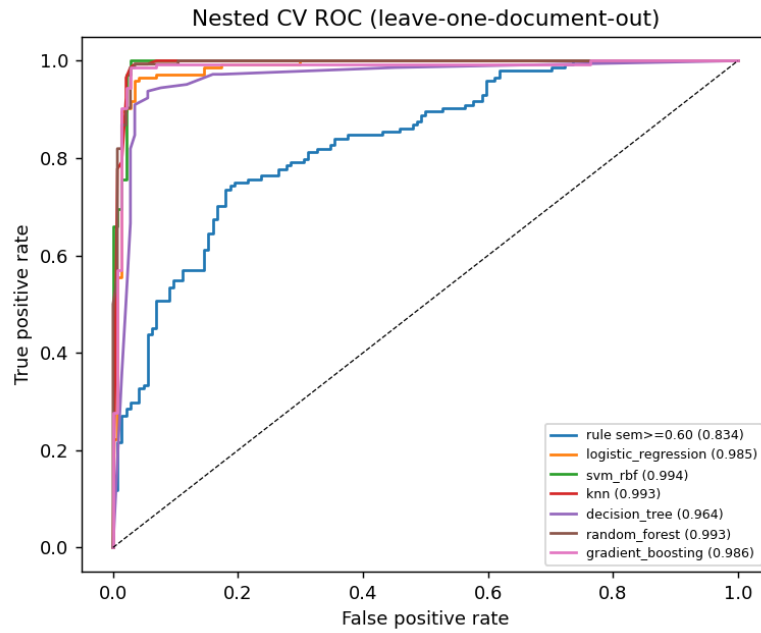


Figure 10: Nested-CV ROC curves for all models and the hand-set rule.

8.3 Results on unseen documents

	Accuracy	Precision	Recall	F1	ROC-AUC	Brier
Trained model	0.9625	0.9664	0.9583	0.9623	0.9938	0.0325
Hand-set rule ($\text{sem_top1} \geq 0.60$)	0.6792	0.6287	0.875	0.7317	0.8337	0.215

On 240 questions from 5 unseen documents the model makes 4 false-positive and 5 false-negative decisions. By category the hand-set rule fails on questions about invented entities (8.6% correct) and absent attributes (14.3%); the trained model handles them (100.0% and 88.6%). Probabilities are reasonably calibrated (Brier score above), so the confidence shown to users is meaningful.

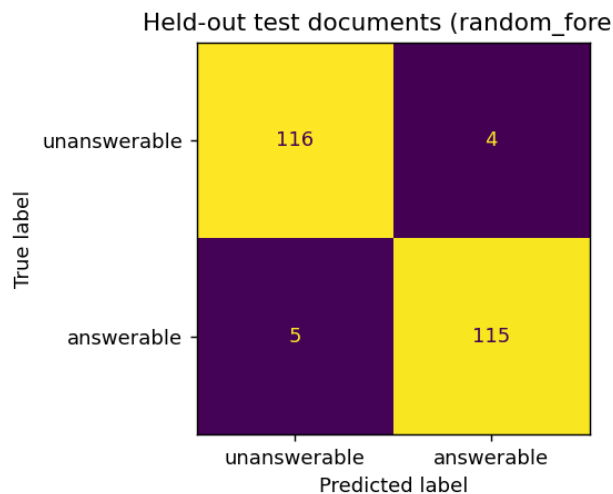


Figure 11: Confusion matrix on unseen documents.

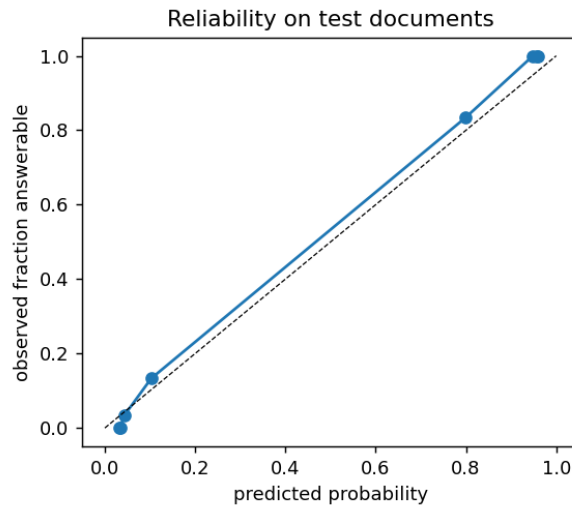


Figure 12: Reliability diagram on unseen documents.

8.4 Independent hand-written questions (end to end)

Question set	System	Overall	Answerable	Refusal
36 questions, 3 held-out domains (never tuned on)	Trained model	0.889	0.875	0.917
	Previous hand-set gate	0.722	1.0	0.167
32 questions, sample HLD (used for error analysis, so no longer independent)	Trained model	0.906	0.885	1.0

The earlier gate answered almost every question, which gave perfect recall but only 16.7% refusal accuracy; its 32/32 on the first sample question set was inflated because the threshold had been calibrated on that same set (data leakage), which is why the independent set was created. The trained model trades a small loss of recall for a large gain in safe refusals.

Error analysis of the independent set:

- **S02** (refused an answerable question): 'What does the AngleSensorManager provide to the rest of the system?'
- **S03** (refused an answerable question): 'Which signal tells the motor drive how much current to apply?'
- **S11** (answered an unanswerable question): 'What is the CAN identifier of IFaultReport?'
- **K07** (refused an answerable question): 'Which component estimates how far away the key is?'

Refusals of answerable questions occur for unusual paraphrases ('how much current to apply' for the AssistCurrent signal) whose wording shares little vocabulary with the document; the wrongly answered question names a real interface and asks for an attribute (CAN identifier) that is absent, the hardest case for a retrieval-based system. Remedies are a larger and more varied training set and a verification step that checks whether the asked-for attribute appears in the cited evidence.

8.5 Entity extraction

Held-out extraction (3 unseen domains): precision 1.00, recall 1.00, F1 1.00 for every entity type. See Section 7.3 for the before/after analysis.

8.6 Validation rules

Across the 9 synthetic documents, 36 of 36 seeded defects were detected (recall 1.00; by kind {'wrong_direction': '9/9', 'missing_dependency': '9/9', 'undefined_interface_on_port': '9/9', 'signal_endpoint_mismatch': '9/9'}), with 0 findings on the clean versions and 0 findings on the defective versions that were not explained by a defect. Caveat: the defect kinds were chosen to match the rules, so the recall confirms correct implementation, not coverage of all possible HLD inconsistencies.

8.7 Optimisation summary

What was optimised	How	Effect (held-out)
Answerability model	Six families, nested grouped CV, grid search, sigmoid calibration, threshold on out-of-fold F1	F1 0.73 to 0.96; AUC 0.83 to 0.99
Retrieval fusion weight	Grid on development documents	MRR 0.837 to 0.869
Stop-word list	Error analysis on refused answerable questions	Fewer spurious uncovered tokens
Extractors	Error analysis on development corpus	F1 0.83 to 1.00

8.8 Limitations of the evaluation

- Synthetic, template-generated data with one document layout; real HLDs vary more.
- Question templates are shared between training and test documents; only documents differ. The hand-written sets are the check against this.
- Gold labels for retrieval are section-level and the documents are small (about 10 chunks), so recall@5 saturates; MRR and recall@1 are the informative metrics.
- The local-LLM answer path, OCR on real scans and user-time savings were not evaluated.
- Test and independent sets are small (240 and 36 questions); differences of a few percentage points are not significant.

9. Implementation

The code base is organised by responsibility: `app/ingestion` (parsing, OCR, metadata), chunking, extraction (pattern and table extractors, normalisation, structured model), graph, rag (tokenizer/BM25, embeddings, vector store, hybrid retriever, answerability model, answerer), validation, comparison, reporting, api, evaluation and a Streamlit frontend. 43 automated tests (unit and integration) cover parsing helpers, extraction, validation rules, retrieval, the answerability artefact, the API, security (roles, project isolation, audit), exports and the version endpoint; the code passes the Ruff linter.

Pipeline. A PDF is validated, parsed per page with PyMuPDF, cleaned, chunked by section, embedded and indexed, and its entities are extracted from patterns and tables and de-duplicated with merged evidence. A typed model of the tables feeds the graph, the 11 validation rules and the revision comparison. At query time BM25 and dense rankings are fused, 12 features are computed, the answerability model decides to answer or refuse, and the answer is composed from graph facts and cited passages with a probability-based confidence.

Rule	Check	Severity
V001	Provider/consumer/port owner not a declared component	critical
V002	Port references an undefined interface	critical
V003	Port direction contradicts component role	critical
V004	Interface endpoint without a matching port	warning
V005	Dependency table missing or contradicting the interface table	warning / critical
V006	Unknown relationship or undefined dependency target	warning / critical
V007	Signal endpoints differ from interface provider/consumer	critical
V008	Signal undefined, unused or without data type	warning
V009	Port name suffix disagrees with direction	info
V010	Component takes part in no interface	info
V011	Functional flow mentions an undefined signal/interface/port	warning

Security and governance. API keys map to roles (viewer, reviewer, admin) and projects; each document has its own vector collection and every route checks role and project; uploads, queries, reviews, exports, comparisons and deletions are written to an audit log; the reviewer recorded on a finding is the authenticated user.

10. Deployment and MLOps basics

- **Model versioning and registry:** each trained model is stored under `models/answerability/v1.0.0` (and current) with metadata: algorithm, hyperparameters, threshold, feature list, training and test documents, dataset hash (3b3dff5e5a71c842), library version, git commit (fc73bae), seed and metrics. The service refuses to load a model whose feature list does not match the code.
- **Experiment tracking:** every training and retrieval experiment appends a JSON record (run id, timestamp, git commit, parameters, metrics, artefact paths) to `data/evaluation/runs.jsonl`.
- **Reproducibility:** fixed seeds, deterministic dataset generation, a dataset hash, pinned dependencies (`requirements-lock.txt`) and single-command scripts for every experiment (appendix).
- **Service traceability:** `GET /version` reports the git commit, embedding model, answerability model version, threshold, training documents and test F1 of the running service.

- **Docker:** one image with the embedding model baked in, docker-compose for API + UI with persistent volumes, and a single-container mode for one-port hosts (Render blueprint included). The build was verified, including login-enabled operation.
- **CI (simulated CD):** a GitHub Actions workflow runs lint, tests, the extraction/validation/Q&A evaluations and a Docker build with a health and version smoke test on every push and pull request.

11. Responsible AI and deployment ethics

- **Safety risk:** a wrong architecture answer could mislead a design review. Mitigations: answers use only retrieved evidence and graph facts, every answer cites section and page, a trained classifier refuses questions the document cannot answer, confidence is a calibrated probability, and findings need human accept/reject decisions. The report and UI carry an advisory notice; the system never approves a design.
- **Hallucination:** by default there is no generative model; an optional LLM is restricted to the retrieved evidence and instructed to reply 'not found' otherwise.
- **Privacy and confidentiality:** embeddings, vector store and database are local; the embedding model runs offline; external APIs are disabled by default and must be declared. Only synthetic documents are used.
- **Access control and accountability:** roles, per-project isolation, authenticated reviewer identity and a complete audit log.
- **Bias and fairness:** the data are technical documents, not personal data, so demographic bias does not apply; the relevant bias is *layout and vocabulary bias*: the system was built on documents with a single table layout and automotive vocabulary, and may fail silently on others. This is stated in the limitations, and the intended failure mode is refusal rather than a wrong answer.
- **Prompt injection:** document text is treated as data and delimited in prompts; not red-teamed.
- **Transparency:** model metadata, thresholds and training documents are exposed through /version and stored with the artefact.
- **Environmental/cost:** small CPU models (about 130 MB embedding model, a classifier of a few hundred KB) keep energy and cost low.

12. Innovation highlights

- A learned, document-grouped answerability model that replaces a brittle similarity threshold and supplies calibrated confidence.
- Evidence-based design decisions: every component choice is backed by a measured comparison on held-out documents, including results that contradicted the initial design (equal-weight hybrid, PCA compression).
- Rule-based validation engine over a typed model of the HLD tables with page-level evidence and a human review workflow; verified on seeded defects.
- Revision comparison with graph-based change-impact analysis; interactive traceability graph and per-component cited reports.
- Leakage-aware evaluation: disjoint development and test documents, an untouched independent hand-written set, and recorded provenance for the model.

13. Screenshots and evidence

Screenshots were captured automatically from the running application (scripts/capture_screenshots.py) using the two sample revisions.

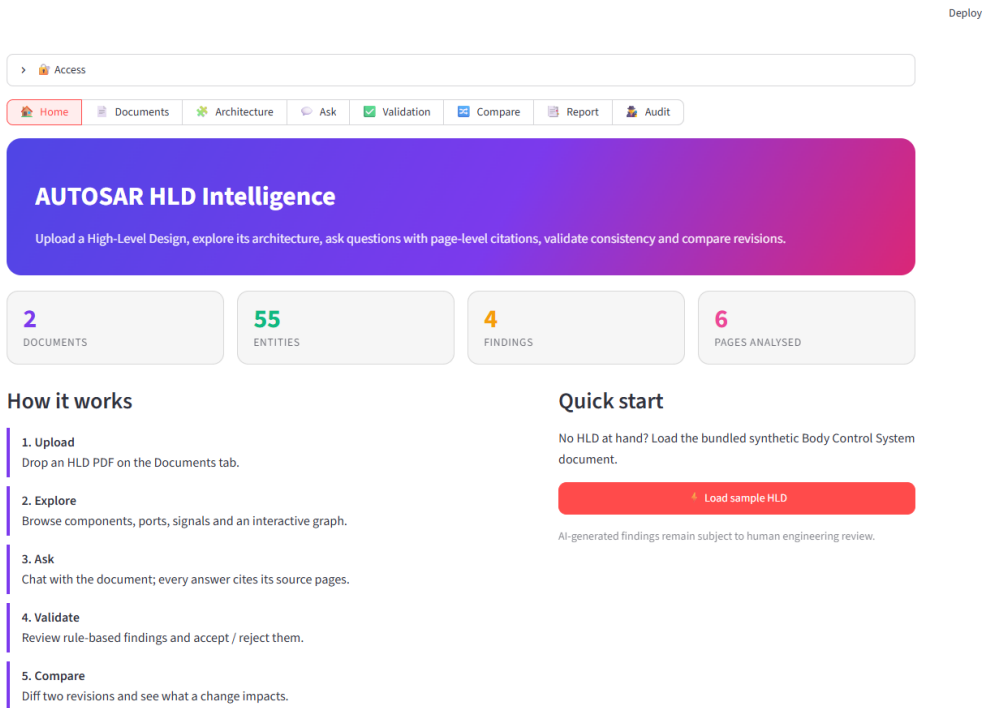


Figure 13: Home page.

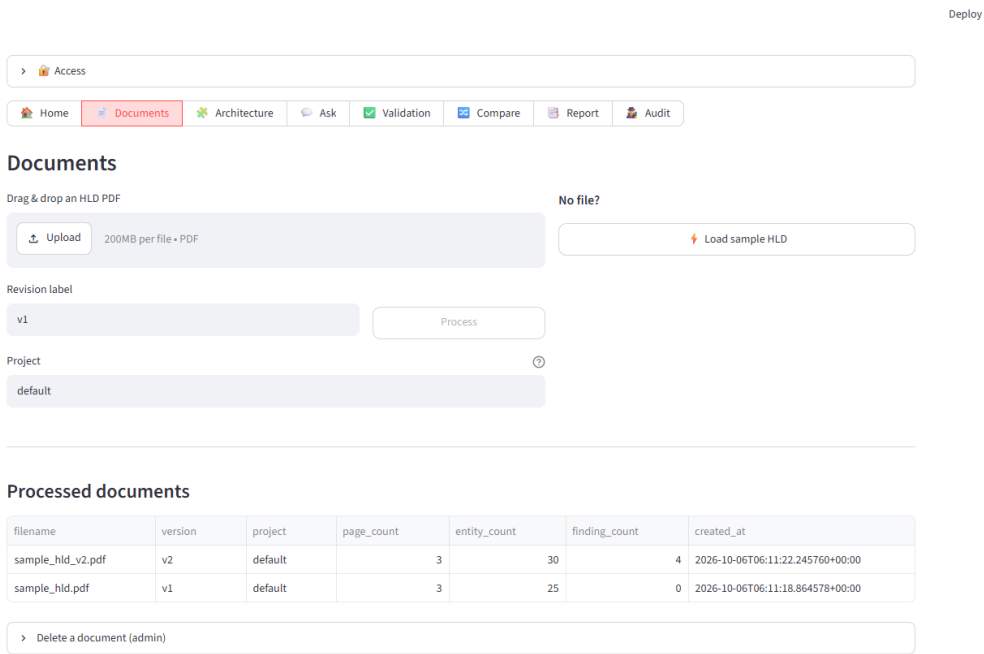


Figure 14: Documents page with project label.

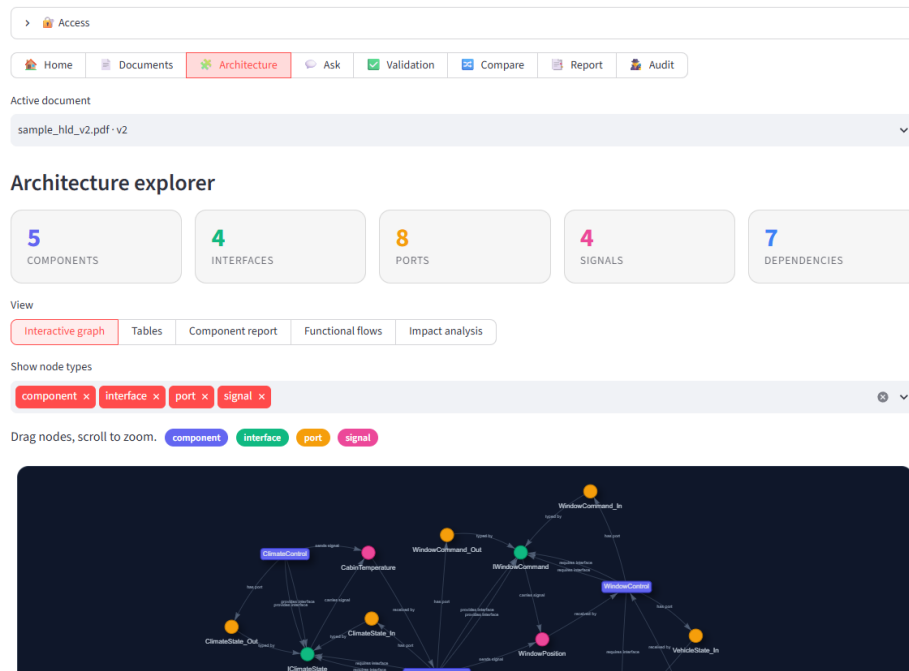


Figure 15: Interactive traceability graph.

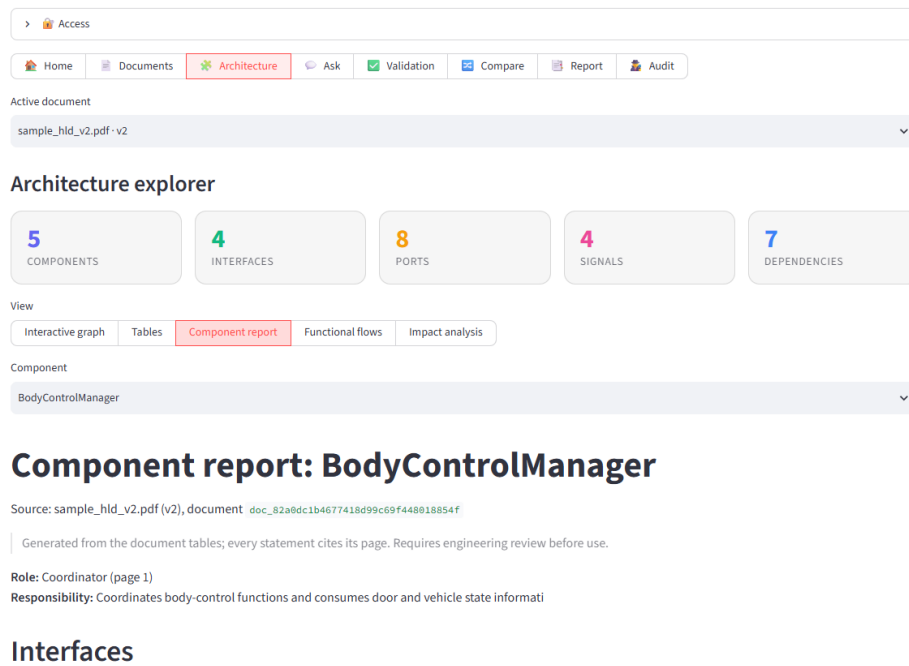


Figure 16: Cited component report.

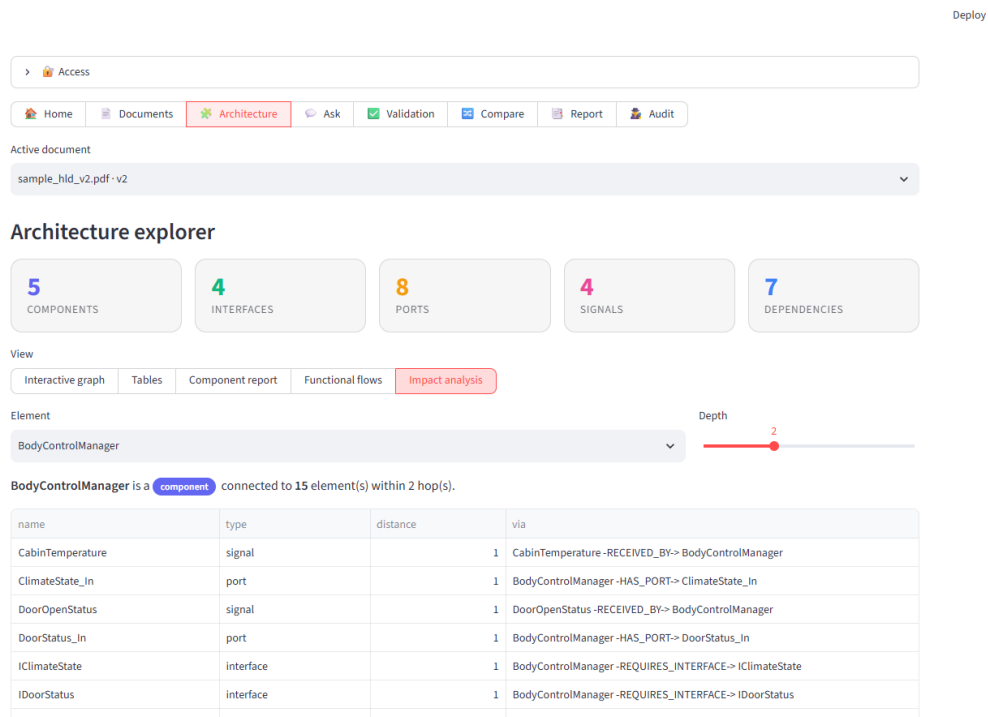


Figure 17: Change-impact explorer.

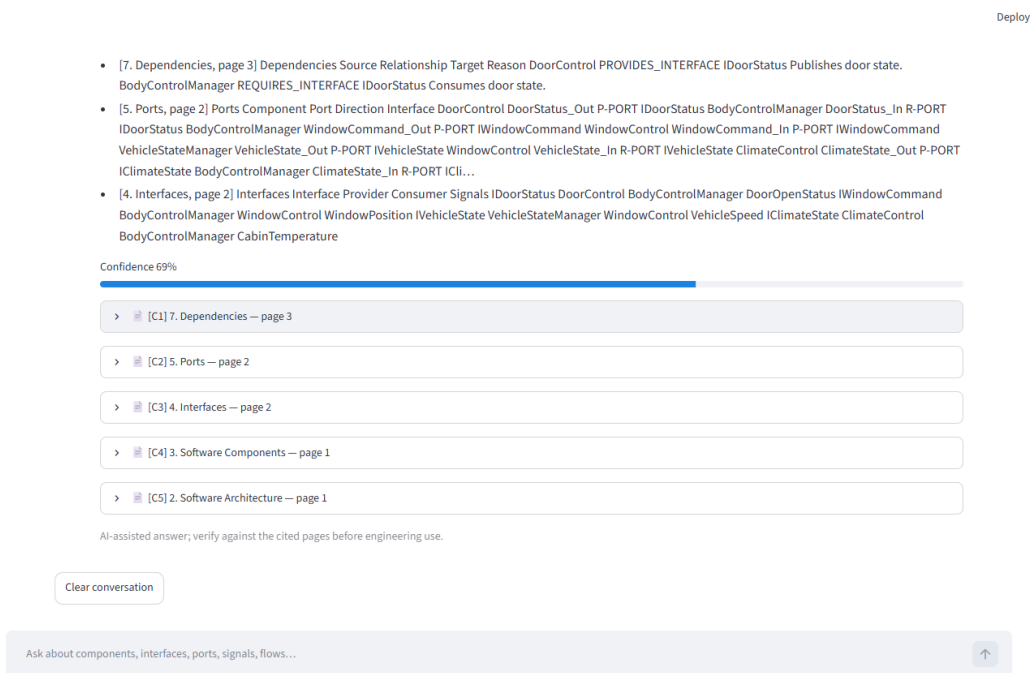


Figure 18: Grounded answer with citations and calibrated confidence.

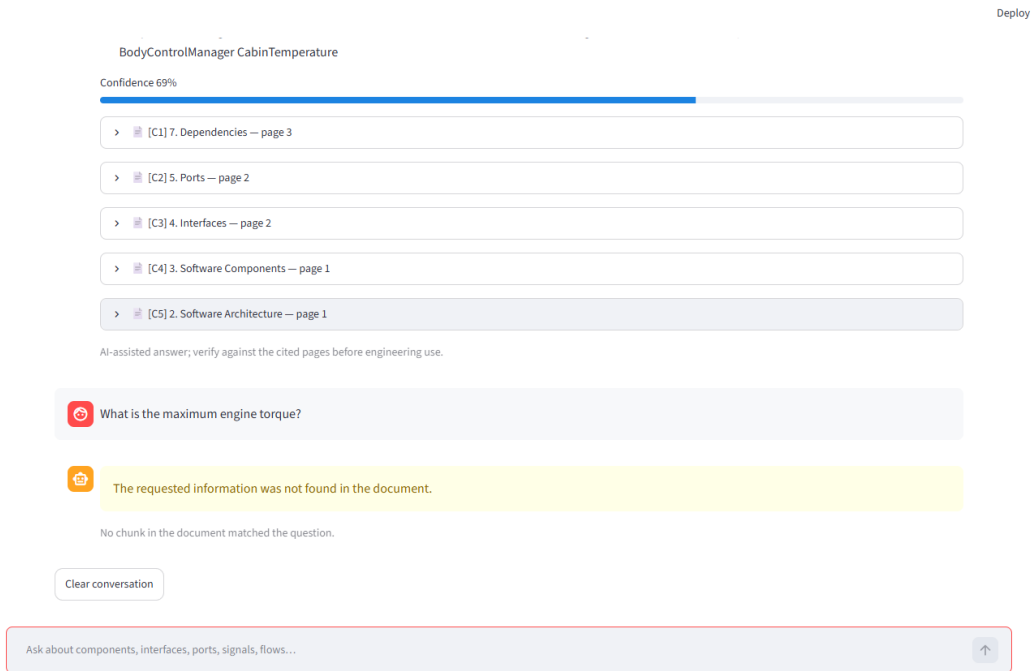


Figure 19: Refusal of an out-of-scope question.

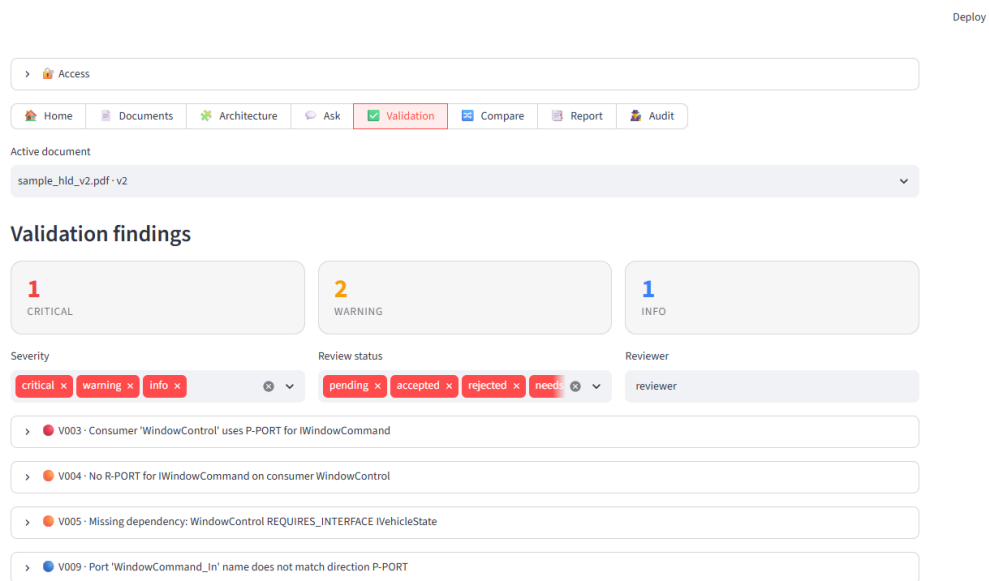


Figure 20: Validation findings and review actions.

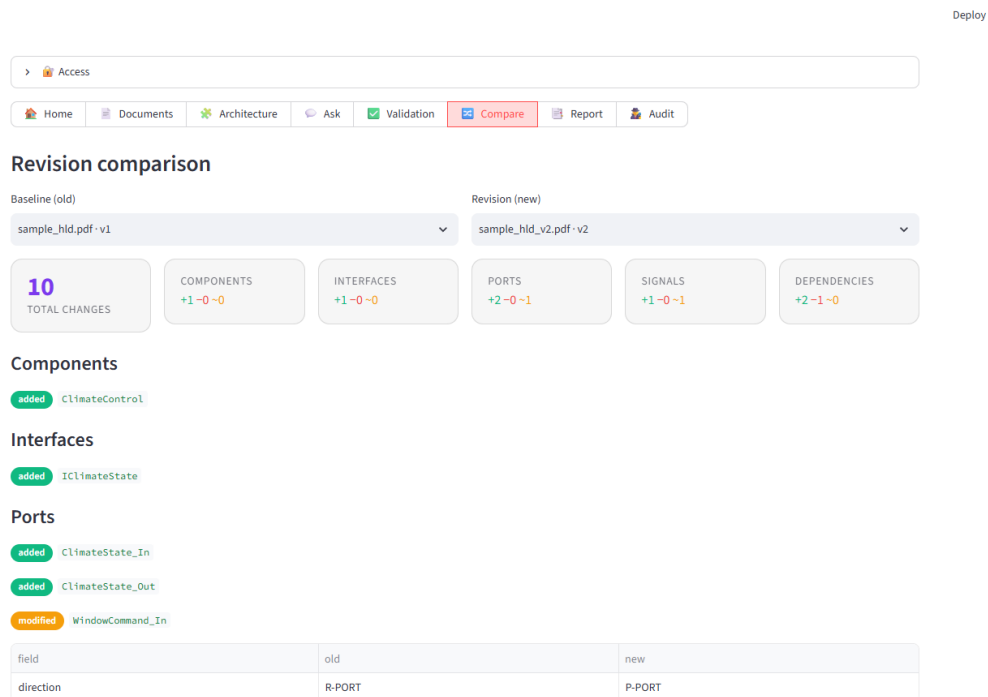


Figure 21: Revision comparison.

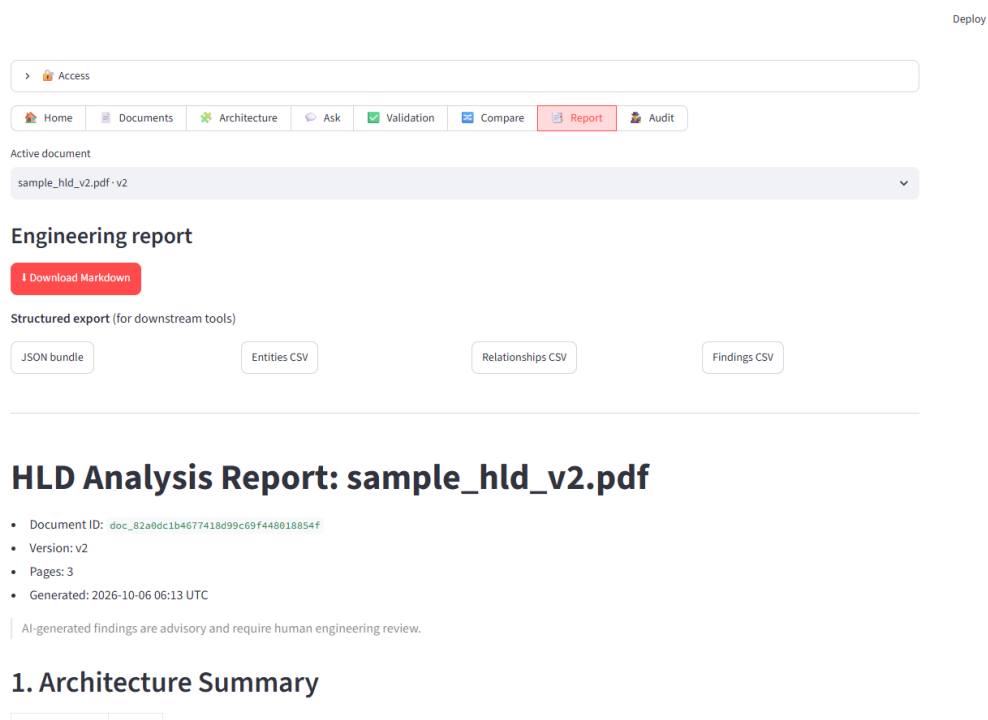


Figure 22: Report page with structured export.

Logs and raw results are included in the submission under Evaluation_Results/ (pytest output, metrics JSON, experiment log, sample outputs with citations).

14. Reflection and conclusions

The strongest lesson was that decisions have to be measured. Three design choices that looked sensible were contradicted by held-out data: a similarity threshold tuned on one document did not transfer (it answered most unanswerable questions), an equal-weight hybrid retriever was worse than the dense model alone, and extractors tuned on one document missed half of the signals in other domains. Grouping cross-validation by document and keeping untouched test documents exposed these problems, and the fixes (a trained answerability model, a tuned fusion weight, table-driven extraction) each improved held-out results.

The main limitation is realism: all data are synthetic and share one layout, the question sets are small, and the system was not evaluated on real HLDs, scanned documents or with a generative model. Next steps are an independent evaluation set from real or differently structured HLDs, table-header mapping for other layouts, a verification step for 'absent attribute' questions, evaluation of a local LLM, a graph database for large documents and enterprise identity management.

Personal reflection (to be written by the student): what you learned, what was difficult, what you would do differently.

Appendix: reproduction commands

```
pip install -r requirements-dev.txt
python scripts/generate_synthetic_hlds.py
pytest
python -m app.evaluation.extraction_eval --label after_fix --out data/evaluation/results
python -m app.evaluation.train_answerability
python -m app.evaluation.retrieval_experiments
python -m app.evaluation.validation_eval --out data/evaluation/results
python -m app.evaluation.eda
python -m app.evaluation.run_eval --out data/evaluation/results
docker compose up --build
```